Symplectic Is Not Enough: Time-Reversal Symmetry in Neural Hamiltonian Proposals

Abstract

A Hamiltonian flow carries two structures: it preserves the symplectic form, and it is time-reversal symmetric, meaning that reversing the momentum, flowing forward, and reversing again inverts the flow. Symplectic neural networks (SympNets) encode the first structure exactly, by architecture. We observe that they do not encode the second, and that this is not a cosmetic gap: when a SympNet is used as the proposal of a Metropolis–Hastings sampler, symplecticity supplies volume preservation but time-reversal symmetry is what supplies the involution required for detailed balance. A trained SympNet violates it by $\mathcal{O}(10^{-1})$, so the resulting chain does not target the intended distribution, and the discrepancy does not vanish with more training data. We give a construction that imposes the missing symmetry exactly rather than approximately. Because every SympNet layer is a shear, it admits a closed-form inverse; composing the network with its momentum-conjugate inverse yields a map that is symplectic and involutive as an algebraic identity in the weights, is consistent (it reduces to the exact flow when the network is exact), and adds no parameters. Empirically the construction moves the reversibility error from 10^{-4}–10^{-2} to machine precision across targets, uniformly in depth, whereas increasing depth alone never fixes it. In a depth-controlled comparison the symmetrized network matches the sampling efficiency of a plain network of twice the depth and four times the parameters. The construction is architecture-agnostic within the SympNet family and extends unchanged to non-separable Hamiltonians.

Keywords: symplectic neural networks, Hamiltonian Monte Carlo, time-reversal symmetry, structure-preserving architectures, geometric integration

1. Introduction

Structure-preserving architectures encode a known invariance of a physical system directly into a hypothesis class, so that the invariance holds for every choice of weights rather than being approached through training. Symplectic neural networks (Jin et al., 2020) are a canonical instance: each layer is a shear on phase space, so the network is a symplectic map by construction and its Jacobian has unit determinant identically.

Hamiltonian dynamics, however, possess more than one structure. Writing $z = (q, p)$ and $R(q, p) = (q, -p)$ for the momentum flip, the exact flow Φ_t of a Hamiltonian that is even in the momentum satisfies both

$$\Phi_t^*(dq \wedge dp) = dq \wedge dp \qquad \text{and} \qquad R \circ \Phi_t \circ R = \Phi_t^{-1}. \tag{1}$$

The first is symplecticity. The second is time-reversal symmetry: reversing the momentum, flowing forward for time t, and reversing the momentum again returns the system to where it began. A SympNet inherits the first property from its architecture. It does not inherit the second, and nothing in the usual training objective — a regression of predicted onto observed states — rewards it.

We argue that this asymmetry matters, and that the setting where it matters most sharply is Markov chain Monte Carlo. Hamiltonian Monte Carlo (Brooks et al., 2011) proposes states by integrating Hamiltonian dynamics and corrects the discretization error with a Metropolis accept–reject step. The correctness of that step rests on exactly the two properties in (1): volume preservation removes the Jacobian factor from the acceptance ratio, and time-reversal symmetry supplies the reverse move required by detailed balance. Surrogate methods that replace the expensive gradient of the log density with a learned one (Li et al., 2019; Dhulipala et al., 2023) retain both properties automatically, because the learned object is placed inside a leapfrog integrator that possesses them. A learned *flow map* is different: it replaces the integrator, so the network itself must supply the structure.

Our contributions are as follows.

- We show that a SympNet used as an MCMC proposal satisfies volume preservation but violates involutivity, and that the violation is substantial in practice ($\mathcal{O}(10^{-1})$ at initialization, 10^{-4}–10^{0} after training). The resulting sampler is not asymptotically exact, and no amount of training data repairs this.
- We give a construction, $\Psi_t = R \circ \Phi_{t/2}^{-1} \circ R \circ \Phi_{t/2}$, that restores the missing symmetry exactly. We prove it is symplectic and involutive for arbitrary weights (Proposition 1) and consistent (Lemma 2), and we give closed-form inverses for the LA- and G-SympNet layers that make it exact and cheap.
- We verify the construction numerically and, in a depth-controlled comparison, separate what the symmetry contributes from what its additional effective depth contributes. Reversibility is restored uniformly in depth, a structural effect; sampling efficiency improves in the weaker sense of parameter efficiency, and we report this distinction explicitly.

2. Background

2.1. Validity of a deterministic Metropolis proposal

Let $\pi(z) \propto e^{-H(z)}$ on $\mathbb{R}^{2D}$ with $H \circ R = H$, and consider a Metropolis–Hastings scheme that proposes $z' = T(z)$ deterministically and accepts with probability $\min\{1, e^{H(z)-H(Tz)}\}$. This leaves π invariant provided

$$\text{(V)} \quad |\det \nabla T(z)| = 1, \tag{2}$$

$$\text{(I)} \quad (R \circ T) \circ (R \circ T) = \mathrm{Id}, \tag{3}$$

that is, volume preservation and involutivity of $R \circ T$ (Green, 1995; Tierney, 1998). Leapfrog integration of any force field depending on position alone satisfies both identically, which is why gradient-learning surrogates remain asymptotically exact: an inaccurate network lowers the acceptance rate but cannot bias the stationary distribution. This robustness is a property of the integrator, not of the learning.

2.2. Symplectic networks

A SympNet composes layers that update either q or p using a function of the other, unmodified half of the state. For the gradient (G) module in "up" mode the layer is $(q, p) \mapsto (q,\ p + \tau\, K^{\top} a\, \sigma(Kq + b))$, and the linear (LA) module composes several such shears with symmetric matrices. Each layer has unit-determinant Jacobian, so (2) holds for any weights. Condition (3) is a statement about the interaction of the map with R, and no layer constrains it.

3. The missing symmetry

3.1. Closed-form inverses

The shear structure that gives SympNets their symplecticity also makes them exactly invertible. In an "up" layer the position q passes through untouched, so the increment added to p can be recomputed from the output and subtracted; the inverse is the same expression with the sign reversed, at identical cost and with no iteration. In the LA module the shear is followed by a translation, so the inverse subtracts the translation first and then undoes the shears in reverse order. Composing layer inverses in reverse order gives Φ_t^{-1} exactly; we measure $\|\Phi_t^{-1}\Phi_t z - z\|_\infty \approx 10^{-15}$ in double precision.

3.2. The symmetrized map

Define

$$\Psi_t \;:=\; R \circ \Phi_{t/2}^{-1} \circ R \circ \Phi_{t/2}. \tag{4}$$

Proposition 1 *If $\Phi_{t/2}$ is invertible then Ψ_t satisfies (3); if $\Phi_{t/2}$ is in addition symplectic then Ψ_t satisfies (2). Consequently the Metropolis–Hastings chain with proposal Ψ_t leaves π invariant for arbitrary network weights.*

Proof Write $\Phi := \Phi_{t/2}$. Since $R^2 = \mathrm{Id}$,

$$R \circ \Psi_t = R \circ R \circ \Phi^{-1} \circ R \circ \Phi = \Phi^{-1} \circ R \circ \Phi,$$

a conjugation of the involution R, hence itself an involution: $(R \circ \Psi_t)^2 = \Phi^{-1}R\Phi\Phi^{-1}R\Phi = \Phi^{-1}R^2\Phi = \mathrm{Id}$, which is (3). For (2), note R is anti-symplectic, reversing the sign of the canonical form; therefore $R \circ \Phi^{-1} \circ R$ is symplectic, and Ψ_t, a composition of two symplectic maps, is symplectic, so $|\det \nabla \Psi_t| = 1$. Neither argument uses any property of Φ beyond invertibility and symplecticity. ■

The half-step in (4) makes the construction consistent: it does not perturb a model that is already correct.

Lemma 2 *If Φ_s is the exact time-s flow of a Hamiltonian with $H \circ R = H$, then $\Psi_t = \Phi_t$.*

Proof By (1), $\Phi_{t/2}^{-1} = R \circ \Phi_{t/2} \circ R$. Substituting into (4) and using $R^2 = \mathrm{Id}$ gives $\Psi_t = \Phi_{t/2} \circ \Phi_{t/2} = \Phi_t$. ■

Finally, the property survives composition: if $R\Psi R = \Psi^{-1}$ then $R\Psi^k R = \Psi^{-k}$, so $(R\Psi^k)^2 = \mathrm{Id}$ for every k, and multi-step trajectories remain valid proposals.

Table 1: Validity conditions for each proposal map, at random initialization ($D = 2$, $L = 5$, $\epsilon = 0.1$, double precision). Volume preservation is measured as $\max |\det \nabla T| - 1$ and involutivity as $\max \|(RT)^2 z - z\|_\infty$.

proposal map	$\lvert\det \nabla T\rvert - 1$	involution error	valid
Exact leapfrog (HMC)	$2 \cdot 10^{-16}$	$3 \cdot 10^{-16}$	yes
Learned force + leapfrog	$3 \cdot 10^{-16}$	$7 \cdot 10^{-16}$	yes
Learned separable H + leapfrog	$7 \cdot 10^{-16}$	$1 \cdot 10^{-15}$	yes
Learned field + leapfrog	$8 \cdot 10^{-16}$	$9 \cdot 10^{-16}$	yes
LA-SympNet (plain)	$1 \cdot 10^{-15}$	$2.4 \cdot 10^{-1}$	**no**
G-SympNet (plain)	$7 \cdot 10^{-16}$	$1.2 \cdot 10^{-1}$	**no**
LA-SympNet (symmetrized)	$2 \cdot 10^{-15}$	$4 \cdot 10^{-16}$	yes
G-SympNet (symmetrized)	$7 \cdot 10^{-16}$	$1 \cdot 10^{-16}$	yes

3.3. Relation to existing work

Equation (4) is the adjoint, or symmetrization, composition of geometric integration (Hairer et al., 2006), in which a method is composed with its adjoint $\Phi_t^* = \Phi_{-t}^{-1}$ to produce a symmetric method; here the adjoint is realized as $R \circ \Phi^{-1} \circ R$. Enforcing exact invertibility so that a learned proposal admits a tractable reverse move is likewise established for neural samplers (Song et al., 2017; Levy et al., 2018). Our claim is not that this machinery is new, but that the SympNet family — which is otherwise an unusually clean example of a structure-preserving architecture — is missing one of the two structures it is usually assumed to capture, and that the omission is consequential wherever the learned flow is used as a proposal. The closed-form layer inverses make the repair exact rather than approximate.

4. Experiments

We train LA- and G-SympNets on warm-up trajectories from Hamiltonian Monte Carlo and use the learned map as a proposal, correcting with the exact Hamiltonian. Targets are a 3-dimensional Gaussian, a 2-dimensional normal-gamma posterior, and a 30-dimensional Gaussian. All models are trained under an identical protocol (Adam, initial learning rate 10^{-2}, learning rate halved on plateau, early stopping, best-iterate restoration), so that comparisons reflect the architecture rather than optimizer tuning.

Structural validity. Table 1 evaluates (2) and (3) directly for each proposal map, using randomly initialized weights to emphasize that these are properties of the construction and not of training. Every integrator-based surrogate satisfies both conditions to machine precision. Plain SympNets satisfy volume preservation but violate involutivity by $\mathcal{O}(10^{-1})$. The symmetrized versions satisfy both.

Disentangling symmetry from depth. The map (4) evaluates the network twice, so a symmetrized network of n blocks has the same *effective depth* as a plain network of $2n$ blocks while carrying half the parameters. Comparing Ψ against a plain network of equal

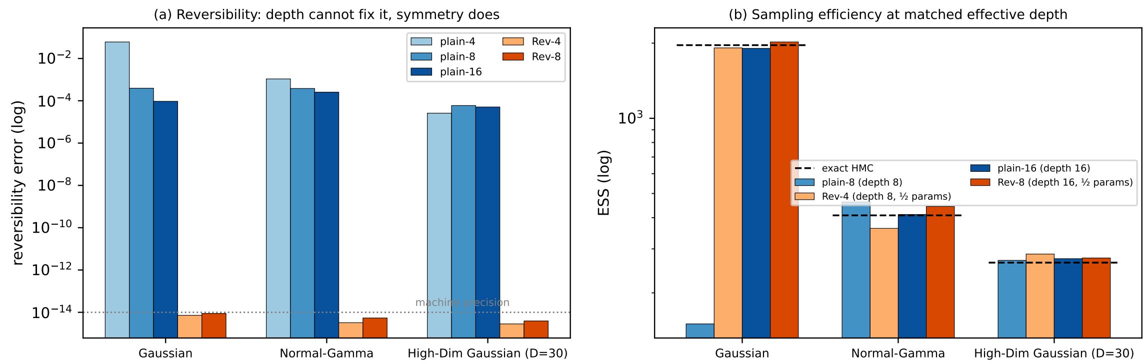

Figure 1: Depth-controlled comparison. Rev-n and plain-$2n$ have equal effective depth; Rev-n has half the parameters. (a) Reversibility error: symmetrization restores machine precision uniformly in depth, while depth alone never does. (b) Effective sample size: the symmetrized model matches a plain model of twice the depth and four times the parameters, and matches exact HMC (dashed).

block count would therefore confound the symmetry with extra depth. Figure 1 compares at matched effective depth.

The two panels separate cleanly. Reversibility (panel a) is restored to machine precision by symmetrization at every depth, and is *not* restored by depth: a plain 16-block network remains at $\sim 10^{-4}$, four orders of magnitude short of even the shallowest symmetrized model. This is the structural effect, and it is unambiguous.

Sampling efficiency (panel b) tells a more modest story, and we state it plainly. Much of the effective sample size gained by symmetrization can also be obtained by simply making the plain network deeper: on the Gaussian target a plain 8-block network attains an ESS of 150, a plain 16-block network 1906, and the symmetrized 4-block network 1914. What survives the control is a parameter-efficiency claim rather than a sampling-quality claim: the symmetrized 4-block model matches the plain 16-block model using a quarter of the parameters, and matches exact HMC (1963). On the remaining two targets the differences in ESS lie within run-to-run variation. We also note that the plain models are non-monotone in depth — 483, 150, 1906 for 4, 8, 16 blocks — while the symmetrized models are stable across depth (1914, 2022), which is consistent with the symmetry acting as a constraint that removes a direction of variation the true solution never uses.

Non-separable Hamiltonians. Because SympNets approximate arbitrary symplectic maps, neither the architecture nor Proposition 1 requires the Hamiltonian to be separable; only the source of the training trajectories changes. We train on Riemannian-manifold HMC trajectories (Betancourt, 2013) generated with the explicit integrator of Tao (2016), recording the full phase-space field $(\partial_p H, -\partial_q H)$, which that integrator already forms at every stage. The symmetrized model attains reversibility of 10^{-13} against 10^0 for the plain model, matching the separable case.

5. Discussion

The observation we wish to emphasize is structural rather than empirical. When an architecture is described as structure-preserving it is worth asking which structures, and whether the downstream use requires others. SympNets preserve the symplectic form exactly and are frequently adopted for that reason; but the flows they are meant to model are also time-reversal symmetric, and that second symmetry is the one an accept–reject correction actually consumes. Because it can be imposed exactly by composition, at no cost in parameters and with no loss of consistency, there is little reason to leave it to training.

Our depth-controlled comparison also carries a methodological caution. The symmetrized map evaluates its network twice, and a comparison that matches block count rather than effective depth would have attributed to the symmetry an improvement largely explained by depth. We recommend that comparisons in this family control for effective depth, and we report the resulting weaker efficiency claim rather than the stronger uncontrolled one.

Limitations are worth stating. Our targets are low-dimensional, and the efficiency benefit we can defend is parameter efficiency rather than improved sampling; establishing the latter would require higher-dimensional posteriors and a comparison against adaptive baselines such as NUTS (Hoffman and Gelman, 2011). Separately, effective sample size alone cannot validate a surrogate sampler: a map with a large involution defect can produce sample paths with low autocorrelation while targeting the wrong distribution, so structural diagnostics of the kind in Table 1 should accompany it.

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